CATERPILLAR®

BEST PRACTICE SERIES

# Preventive Maintenance (Strategic)

Application | **Maintenance** | **Site Management** | Component Rebuild | Safety | MARC Management


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## 1.0 Introduction

Preventive Maintenance (PM) is critical to the success of any maintenance and repair system. It is one of the foundation elements and, as such should be considered a focal point in the overall equipment maintenance management strategy. The goals of PM are to maintain equipment in optimum operating condition, to reduce the risk of failure and subsequent unscheduled equipment repairs, and to optimize equipment owning and operating costs.

In order to meet these goals the well-designed PM process should consist of multiple characteristics. Obviously, as the name implies PM must first be preventive in nature, that is some of the key ingredients of PM are time-based, fixed-interval activities that are planned and scheduled based on manufacturers' recommendations. These preventive activities are focused primarily on eliminating excessive wear in components and systems to ensure that optimum component lives are achieved and that equipment runs in peak operating condition. Routine service activities such as fluid and filter changes are examples of the preventive aspects of PM.

To be completely effective Preventive Maintenance must also have predictive qualities, i.e. periodic inspection, measurement and non-destructive test activities should assist the maintenance organization in the determination of when adjustments or component repair / replacement are required. Predictive maintenance activities are condition-based relative to some predetermined set of criteria and should focus on defect detection and failure avoidance to reduce the risk of equipment breakdown. Predictive activities tend to be more complex relative to their preventive counterparts and range from the most basic visual or "seat of the pants" observations, "special tests" involving specialized instrumentation for pressure and temperature measurements, fluids management and analysis, and on-board electronic system data analysis.

Lastly, since Preventive Maintenance is the most frequent and predictable of planned maintenance activities, it presents a convenient "window of opportunity" for corrective repairs to address problems identified through condition monitoring. Although it may appear to be somewhat reactive, adding this corrective quality to your Preventive Maintenance process is crucial since it supports the goals of equipment reliability by insuring that defects detected during condition monitoring can be planned, scheduled and executed on a before failure basis. In other words, it is much more desirable to react to symptoms and small problems on a planned basis during PM than to attempt to react to failures which by their nature tend to be unexpected and unplanned.

This Best Practice relates specifically to the Preventive Maintenance process applied to large Off Highway Trucks performed in a shop environment and is covered under version II of Site Assessment items 2.9.1 - 2.9.9.

## 2.0 Best Practice Description

Preventive Maintenance consists of a logical sequence of steps or sub-processes that typically include:

- ✓ Preliminary preparation & organization,
- ✓ Initial PM activities,
- ✓ Core PM activities,
- ✓ PM decision criteria & audit process, and
- ✓ Finalizing the PM.

The following is a brief description of each of those steps:

- Preliminary preparation & organization

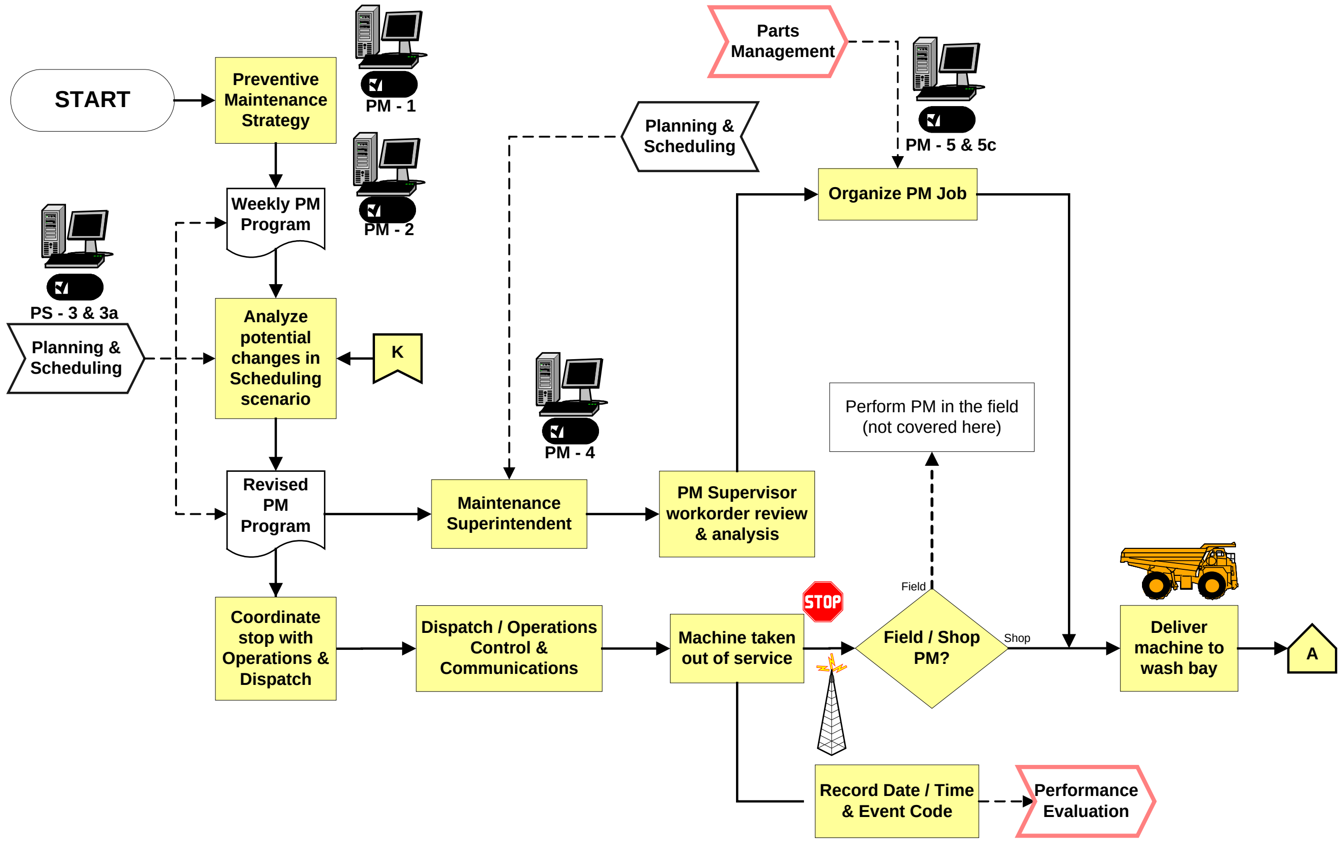


Figure #1 - Preliminary Preparation & Organization Process Flow

The Preventive Maintenance program is generated by Planning and Scheduling and agreed upon by the Operations / Production Department. The program forecast is developed a week in advance and must be sufficiently flexible to accommodate adjustments to reflect deviations in hour accumulation projections. The program takes each of the following into account:

- ✓ PM service interval targets,
- ✓ Resource availability (manpower, bay space, parts, tooling, etc.),
- ✓ Use of the stop as a “Window of Opportunity” to perform other tasks (backlogged and other scheduled repairs),
- ✓ Impact on fleet availability guarantee / commitment,
- ✓ Mine production requirements (status of the production plan).

Once the plan moves into action the PM service package is delivered to the Maintenance Department for review. The PM service package includes the work orders for PM and backlogged repairs to be performed during PM, all PM checklists, Gantt charts, parts consists lists for PM and repairs to be performed during the PM, special instructions, etc. It is passed to the Supervisor or individual in charge of the PM bay. This enables that person to organize the job, i.e. verify that the parts required for PM and any other work on the schedule have arrived and are distributed such that the work is performed in the most efficient manner possible, review the sequence of events (Gantt chart) to confirm that it is logical, assign manpower, and distribute the workload among the staff.

Parts and parts kits are to be delivered and held in a secure location in the PM bay, the day prior to the scheduled PM date. It is the responsibility of the individual in charge of PM activity to verify that the parts required for PM and any other work on the schedule have been delivered to the PM bay, are complete (nothing is missing), and are distributed / organized such that the work can be performed in the most efficient manner possible. Parts are organized as follows:

- ✓ PM parts: mandatory, exchange and optional parts kits,
- ✓ Backlog and scheduled repair parts.

- <u>Initial PM Activities</u>

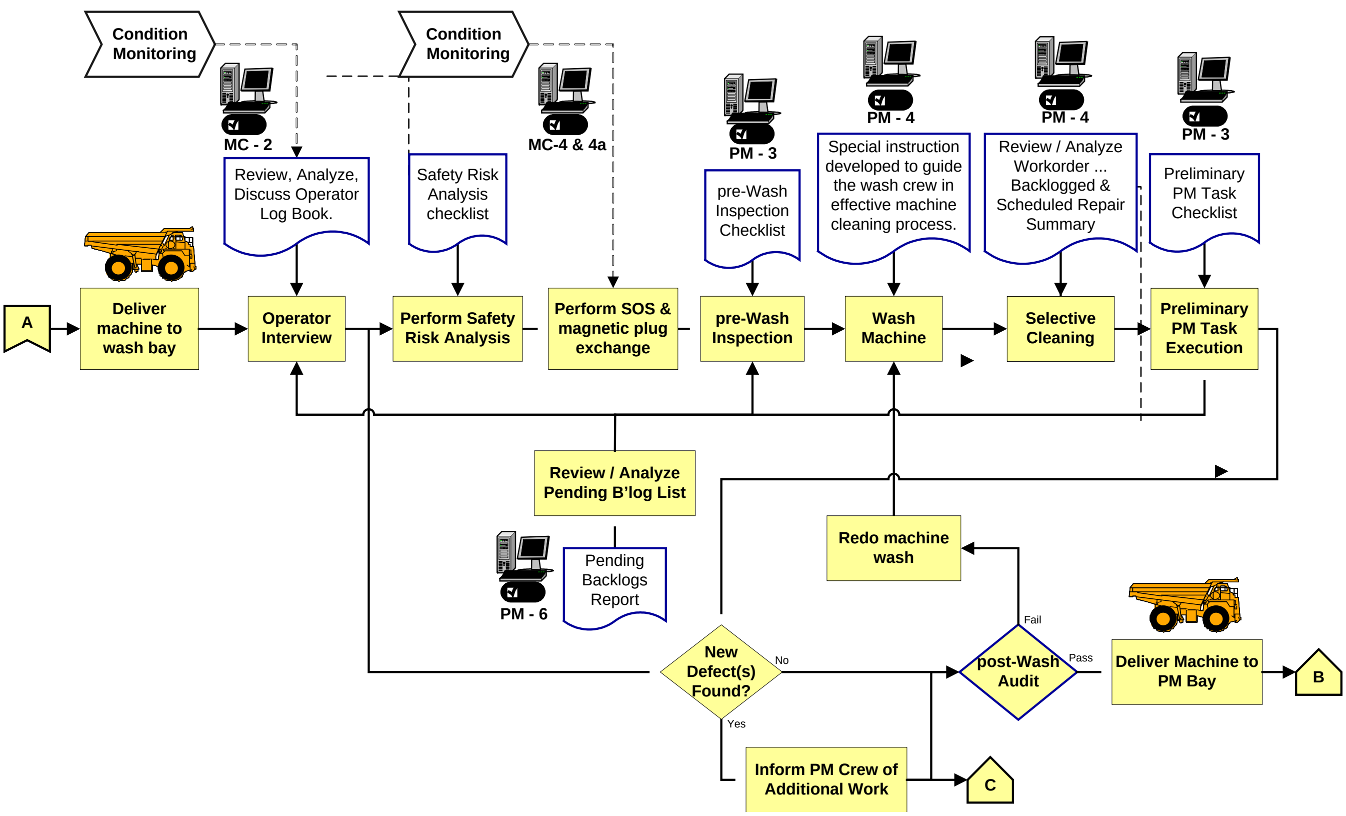


Figure #2 – Initial Preventive Maintenance Process Flow

Once the equipment has been taken out of service and delivered to the wash bay, the Operator Log Book should be reviewed and discussed between the operator and the Technicians responsible for performing the PM. It is important to note that information received from operators tends to be relatively low-tech, visual, symptom-oriented, and "seat-of-the-pants" observations thus PM Technicians must be sufficiently skilled and experienced to interpret operator input and translate the feedback into action.

Before beginning any work on the equipment a Safety Risk Analysis must be performed to minimize any chance of personal injury. Once completed, fluids sampling and magnetic plug exchange should be performed in order to obtain a valid sample before debris can settle out of the oil. Clearly a pre-PM inspection that includes fluid sampling (SOS), magnetic plug inspections, VIMS and other on-board system downloads performed several days in advance of

the PM is preferable however circumstances may prohibit this from taking place. Therefore, if site-specific conditions or policies prevent the pre-PM inspection from taking place, it is important that this activity takes place as soon as possible in the actual PM process.
Washing can sometimes remove evidence of a defect or pending problem. Therefore a visual, walk-around inspection may help identify the source of leaks … oil, fuel, coolant, etc. Other system performance tests / checks are better performed outside the shop or PM bay. Those include:

- ✓ Loose, missing, broken bolts, fasteners, clamps, etc.,
- ✓ Accident damage, i.e. bent sheet metal, broken parts, etc.,
- ✓ Air system performance,
- ✓ Air compressor condition,
- ✓ Brake performance,
- ✓ Parking brake performance,
- ✓ Engine performance (@ torque converter stall),
- ✓ Hoist system performance,
- ✓ Traction control system performance, etc.

Results from these preliminary inspections should be compared to the pending backlog summary and any new defects reported to the PM Supervisor for attention later on in the process.

Equipment cleanliness is critical to contamination control and sound overall maintenance practices. The well-run PM operation should include training for the wash crew instructing them on detailed procedures for the proper execution of this important task. In addition, procedures should be developed warning the crew to avoid directing water spray on areas such as engine exhaust stacks, electrical / electronic components (alternators, etc.), electrical connectors, breathers, etc. To promote efficiency / ease of repair, contamination control and shop housekeeping, the list of all repairs to be performed in parallel with the PM (scheduled or backlogged) should be reviewed to determine if those areas of the machine would benefit from additional, selective cleaning. The source document is any workorder related to scheduled or backlogged repairs. Equipment cleanliness should be audited and, if it does not meet required standards, any deviations attended to prior to moving on to the PM bay.

Since safety is fundamental to any successful PM operation, the following tasks are best performed after the machine has been washed but before it has been moved into the PM bay:

- ✓ Hoist cylinder mounts (pins & bearings),
- ✓ Steering linkage (ballstuds, pins & bearings),
- ✓ Steering system cycle time,
- ✓ A-frame bearing wear,
- ✓ Body pivot pin & bushing,
- ✓ Body safety cable,
- ✓ Suspension cylinder pressures,

Here again, results from these preliminary tests and inspections should be compared to the pending backlog summary and any new defects reported to the PM Supervisor for attention later on in the process.

- Core PM Activities

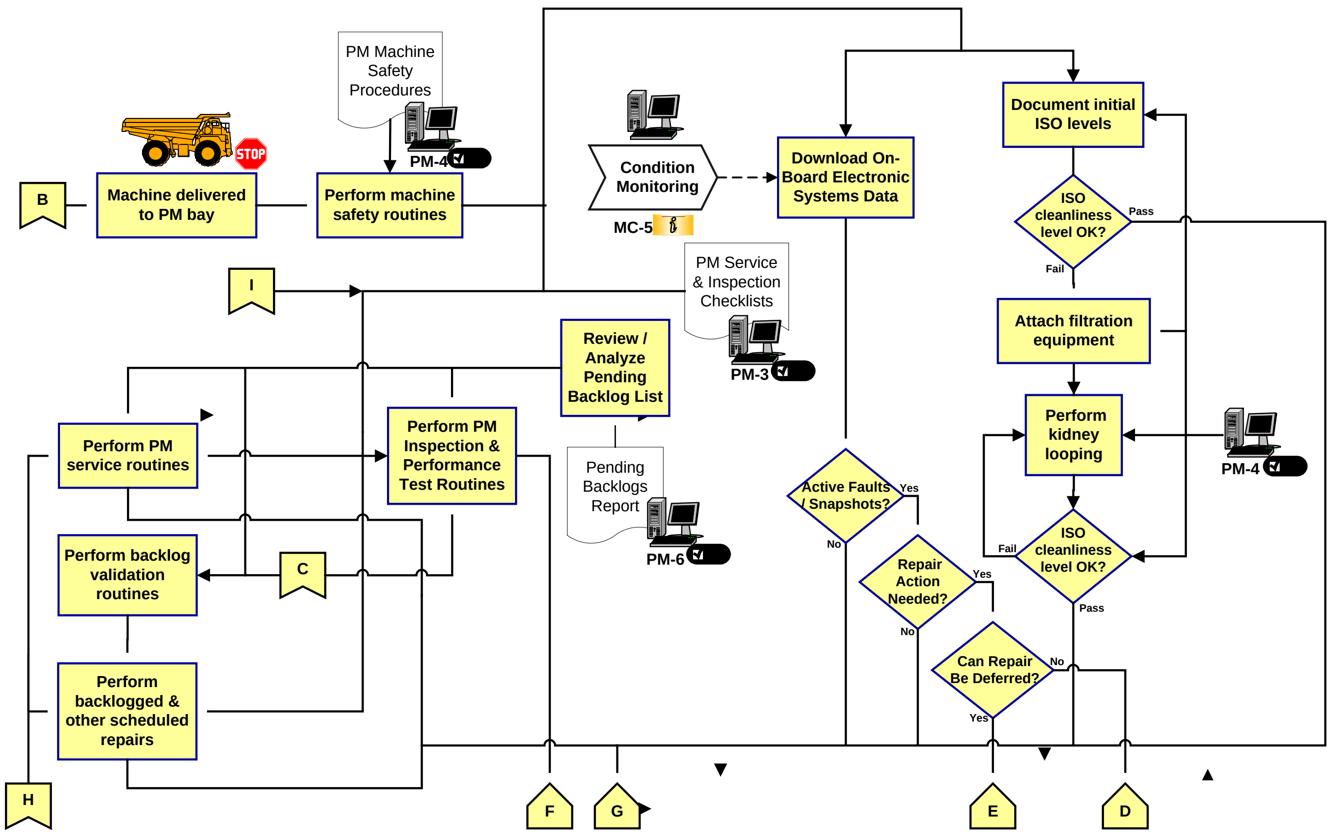


Figure #3 – Core Preventive Maintenance Process Flow

The following safety procedures must be complied with prior to beginning work in the PM bay:

- ✓ Turn master switch to the off position,
- ✓ Apply the parking brake,
- ✓ Place the wheel chocks in position,
- ✓ Apply lock-out tags and other warning devices as required,
- ✓ Raise dump body & attach safety cable.

Fluids cleanliness is a basic step in achieving optimum component lives. As such, the PM process should begin with checks of ISO cleanliness levels and system / component flushing. Once the initial particle count readings have been recorded, the "kidney looping" hardware should be installed and the process performed on the following components and systems:

- ✓ Differential housing & rear wheel groups,
- ✓ Transmission & torque converter groups,
- ✓ Steering system,
- ✓ Hydraulic system,

Final particle count readings should also be recorded as well as the time required to achieve oil cleanliness specifications. NOTE: If the oil in any of these systems is to be changed as part of the PM, the initial (pre-oil change) reading should be documented and the kidney looping process carried out after the oil has been replaced.

On-board electronic systems data may be downloaded concurrent with the kidney looping process for comparison to baseline reference data, predetermined acceptability criteria, and data collected previously to predict pending problems or identify changes in equipment status. These on-board system checks will assist in determining future work plans and revisions to the maintenance plan or strategy. The analyses should focus on the following areas:

- ✓ Maintenance … events / problems with the machine or a system on the machine, component operation / performance, component life management / prediction / optimization, failure analyses …
- ✓ Production … payload management, cycle times / distances, production report summaries / analyses …
- ✓ Operations … changes in application severity, compliance with system operating parameters, haul road management, machine performance...

If active faults or snapshots are discovered, decisions should be taken as to whether corrective actions are required and, if so, if they should be addressed immediately or deferred to the backlog list. Just as in SOS and magnetic plug inspections, it is preferable to download VIMS and other on-board systems several days in advance of the PM as part of a pre-PM inspection. It is mentioned here only because site-specific circumstances, conditions or policies may prevent the pre-PM inspection from taking place.

The remaining PM activities should be performed in the sequence defined by the Gantt chart developed by Planning and provided in the PM Service Package. Preventive Maintenance service tasks, e.g. drain / refill fluid compartments, replace filters, exchange filter canisters, screens, breathers, magnets, replace seals, etc., are to be performed per the defined service interval on each of the following components and systems:

- ✓ Engine including fuel system, fuel tank & air intake system,
- ✓ Fan drive hub,
- ✓ Transmission, Torque converter, & Differential,
- ✓ Rear wheel groups, Brake system including brake wear measurement,
- ✓ Front wheel hubs,
- ✓ Steering & Hoist systems.

PM inspection and performance test routines are to be performed per schedule on each of the following components and systems:

- ✓ Engine including fuel system & fuel tank,
- ✓ Transmission,
- ✓ Torque converter,
- ✓ Differential,
- ✓ Rear wheel groups,
- ✓ Brake systems,
- ✓ Front wheel hubs,
- ✓ Steering system,
- ✓ Hoist system.

Findings should be referenced to manufacturers' specifications and acceptability criteria. All results should be documented immediately so as not to be lost or forgotten.

Backlog validation routines are designed to confirm that issues entered into the backlog system are accurate and to assess potential defects identified during the preliminary (wash bay) inspection. The Pending Backlog Report is one of several "operational reports"; it is a "must" for any person involved in the PM inspection or who will enter a Backlog Request. The primary objectives of this report are:

- ✓ Facilitate the detection routines by letting the originator know what is already in the records,
- ✓ Avoid duplication of entries,
- ✓ Increase the use of any "Window of Opportunity" to execute Backlogs in parallel with PM.

Backlogged and other scheduled repairs can be executed in the PM bay or in the shop repair bay as indicated by the Gantt chart for the PM.

- PM Decision Criteria & Audit Process

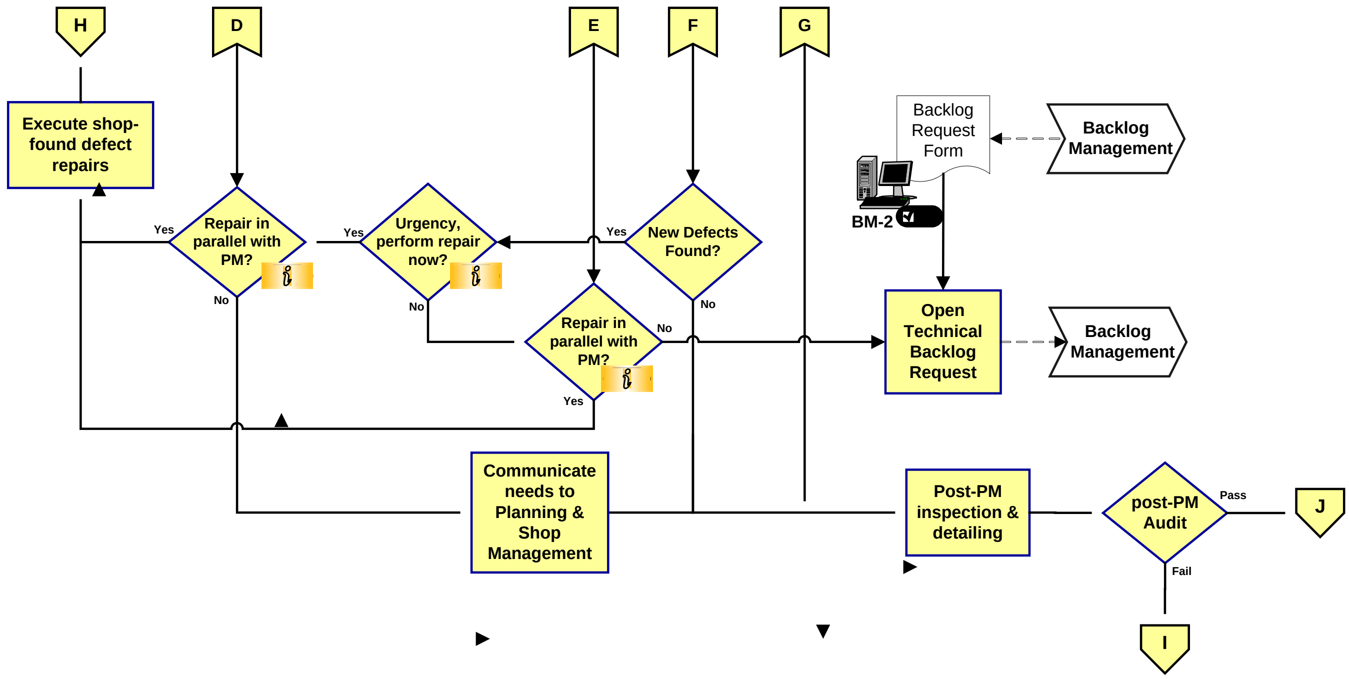


Figure #4 - Decision Criteria & Audit Process Flow

Shop-found defects, problems identified during the preliminary (wash bay) inspection, PM inspections, and/or performance test routines, are the most difficult to deal with. The inability to plan ahead complicates the issue. Proper handling of shop-found defects is a critical aspect of any equipment management strategy.

As such, several things need to be in place to streamline the decision-making process relative to shop-found defects. The PM Team should be trained to distinguish urgent from non-urgent defects. This training can be supplemented through the development, publication, and communication of clear guidelines that provide direction in the determination of the criticality of shop-found defects. An example of repair urgency assessment guidelines may read as follows:

- ✓ The repair of shop-found defects should be considered urgent and executed immediately if they meet the following criteria:

- Defects that have safety implications, e.g. problems with the steering, brake, lighting, critical warning systems, etc., or
- Defects that carry significant risk in terms of contingent damage, repair cost premiums, or increased time to repair as a result of running to failure versus repair-before-failure, e.g. structures, most major components, etc.

Other critical ingredients for success include:

- ✓ PM Technicians must be analytical and knowledgeable in safety / risk assessment.
- ✓ PM Technicians must feel empowered to make decisions and act upon them.

If the repair priority of a shop-found defect is determined to be urgent, the following questions should be asked:

- ✓ Will the execution of the repair require staff (specialized or other) that is not readily available?
- ✓ Will the time required to execute the repair in parallel with the PM cause the next scheduled PM to be delayed due to a conflict for PM bay space?
- ✓ Will the repair require parts to be procured that will delay the start of the next scheduled PM?
- ✓ Will the repair require tooling or other resources to be obtained that will delay the start of the next scheduled PM?

If the answer to any of the questions above is “Yes”, the machine should be moved to the main shop and the repair made under the guidelines defined in Repair Management.

If it is determined that the repair priority of a shop-found defect is not urgent, the guidelines below are applicable:

- ✓ Minor defects, i.e. loose bolts, fasteners, fittings, leaks, etc. that can be executed quickly and without the need for parts, special tooling / instructions, etc. should be performed in parallel with the PM provided there is no interference with or delays in the existing PM schedule.
- ✓ Repair of minor defects that will interfere with or delay the existing PM schedule due to the time required to execute the repair, obtain / organize manpower, parts, special tooling / instructions, etc. should be entered into the Backlog Management process.

A fully functional Backlog Management system ensures that details are captured and not lost. As was noted above, if the defect is determined to be non-urgent but cannot be repaired in conjunction with the PM, it must be entered into the Backlog Management system in the form of a backlog request. This record is used to communicate pending repairs to the Planning Department whose responsibility is to plan and schedule the repair in the most efficient, effective manner possible. The following are critical to the success of the Backlog Management system:

- ✓ Avoid duplication … compare new entries with the pending backlog list,
- ✓ Ensure that information provided in the backlog request is as accurate and complete as possible,
- ✓ The percentage of backlogs identified during PM should be minimal.

Executing the PM program as planned can spell the difference between success and failure. The post - PM audit process is the best way to ensure compliance with the PM plan. If properly communicated, the audit results can be used in an ongoing continuous improvement effort to refine the PM process to meet site-specific requirements. The audit process can take one of the following forms:

- ✓ Supervisor sign-off at critical points in the process (preferred method),
- ✓ Random checks after PM completion.

Using post-PM audit results to define PM training opportunities, refine / improve the PM checklists, and identify shortcomings in the PM process will enhance PM quality and effectiveness.

- Finalizing the PM

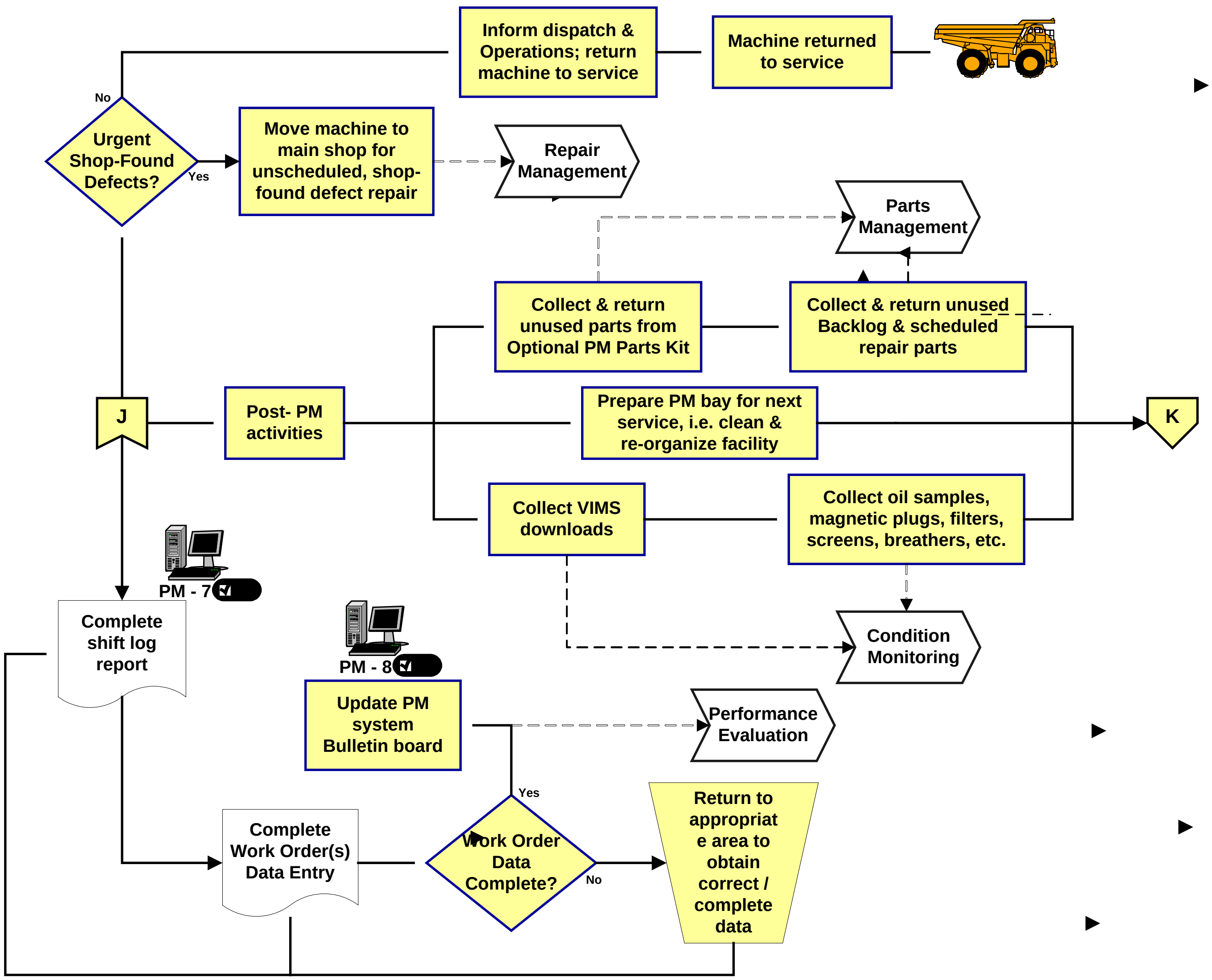


Figure #5 – Finalizing the Preventive Maintenance Process Flow

Once work in the PM bay has been completed, the machine should be moved to the main repair shop to correct any urgent, shop-found defects identified during the PM process. If none were found, the PM Supervisor or individual responsible for PM should advise dispatch and the Operations Department that the machine is available for service.

Many management decisions are based upon knowledge of the workload. If this information is not kept up to date and communicated, management has little idea as to the amount of available manpower at any given point in time. Most mines do a reasonably good job through the use of manual or electronic scheduling boards. It is the responsibility of the individual in charge of PM to inform Planning as soon as a PM is complete. This action takes the machine "off the books" of the Maintenance Department, the PM downtime clock stops running, and the time recorded on the PM workorder.

After the machine has been released from PM, it is the time to correct any shortcomings in the data. The PM Supervisor or individual responsible for PM should audit all paperwork to verify that it is complete and accurate before passing it on to the Planning Department for entry into the machine history archive. Data sources from PM typically include:

- ✓ PM checklists,
- ✓ Work orders for PM, backlogs, and any scheduled or unscheduled repairs executed during the PM process,
- ✓ Backlog requests,
- ✓ Maintenance shift log.

Any deviations or omissions in the results must be corrected before forwarding the data to Planning and the PM team should be advised to ensure that these problems are not repeated in the future.

Information and documentation related to PM should be distributed to the proper personnel as quickly as possible to assure that it is acted upon. This information includes:

- ✓ PM work orders & checklists,
- ✓ Downloaded electronic files,
- ✓ Fluid samples,
- ✓ Magnetic plugs, filters, screens, strainers, etc.

Distribution of PM information results to the appropriate individuals in a timely manner facilitates its use as a source for process performance improvement. PM results and related information can be used to identify problems not only with the equipment but also with the PM execution and Planning & Scheduling processes.

Parts can be a significant cause of delay downtime if they are not managed correctly. Effective PM parts management will reduce maintenance costs and streamline the process by making it more efficient. Parts returns normally associated with the PM process are:

- ✓ Unused parts from optional PM parts kits,
- ✓ Parts from backlog or other scheduled repairs that were not completed during the PM.

In order to be fully effective, an individual should be assigned to be responsible for managing PM parts. It may be necessary to devise a special PM parts return policy since many Parts Departments struggle with parts returns once they have been assigned to a workorder.

Clean, orderly facilities support quality work. Preparing the PM bay for the next scheduled service ensures that facilities are maintained such that future work can be performed as efficiently and effectively as possible. This preparation includes:

- ✓ Cleaning / removing fluids from the floor to eliminate potential work hazards,
- ✓ Returning tooling and equipment to its proper location,

- ✓ Replenishing supplies, e.g. shop rags, cleaning fluids, etc., as required, to support efficient PM execution.

Keys to success in this area include the development of a checklist for procedures to maintain the PM bay in good working order and assigning individual accountability for keeping the PM bay in good condition.

## 3.0 Implementation Steps

- Preventive Maintenance Strategy

Critical steps associated with the implementation of the Preventive Maintenance process begin with the definition and development of the PM strategy. The PM strategy should be documented in writing and must consider and define each of the following elements:

- ✓ PM service interval … typically 250 to 500 hours,
- ✓ The PM plan … define the equipment to be serviced and the scope of work to be performed,
- ✓ Manpower,
- ✓ Training,
- ✓ Infrastructure and resource requirements,
- ✓ PM evaluation measurements / reports … how will PM performance be measured?

- PM Plan Definition

What machines will be covered, how many and which routines will be included in the PM implementation plan, e.g.

- ✓ 20 X 793D Off Highway Trucks,
- ✓ Machine washing,
- ✓ PM service (lubes, oils and filters),
- ✓ Condition Monitoring, e.g.
  - ➢ Oil sampling, inspection of magnetic plugs, screens & filters,
  - ➢ Visual inspections for damage and wear,
  - ➢ System performance tests and adjustment,
  - ➢ On-Board electronic systems data management,
- ✓ Execution of opportunity repairs (backlogged and other repairs).

- Manpower

What human resources are required to implement and execute the Preventive Maintenance plan, i.e. quantity, roles & responsibilities, personal attributes, skills, etc. (Also see Resource Requirements / Human Resources in section 5 .0)

- Training

How can personnel competency be assessed? What training will be needed to develop the skills required to perform the tasks related to Preventive Maintenance?

- Infrastructure and Resource Requirements

What infrastructure and resources will be required to execute the Preventive Maintenance plan?:

- ✓ PM / Wash bays … quantity and physical dimensions? Dedicated facilities?,
- ✓ Special tooling and instrumentation … dedicated to PM?,
- ✓ Parts … PM parts kits, backlogged repair parts, etc.,
- ✓ Machine-specific inspection checklists and forms … “traditional PMP”, “8 Step”, Trakindo “PM Workscopes”, etc.,
- ✓ Well-defined procedures … documented in writing.

(Also see Resource Requirements / Logistical Resources in section 5 .0)

- PM Performance Measurement / Reporting

What defines success in Preventive Maintenance? What performance measures are available that aid in quantifying our efforts in Preventive Maintenance? Once these questions have been answered, the following issues need to be taken into account when developing a plan to quantify Preventive Maintenance performance:

- ✓ Raw input data requirements,
- ✓ Data sources,
- ✓ Data collection frequency and responsibilities,
- ✓ Data flow and management,
- ✓ Calculation methodology,
- ✓ Performance goals and acceptability criteria,
- ✓ Analysis and interpretation roles & responsibilities,
- ✓ Analysis and interpretation guidelines,
- ✓ Information management / archiving,
- ✓ Report formats (templates),
- ✓ Report timing and frequency,
- ✓ Communications / distribution network,
- ✓ Distribution format and media.

Please see the document “Metrics (KPI’s) to Assess Process Performance” for a complete listing and explanation of Preventive Maintenance process performance metrics.

## 4.0 Benefits

Successful implementation of a comprehensive Preventive Maintenance process, i.e. one that includes not only preventive qualities but is also predictive and corrective in nature, will support the goals of most mining organizations as follows:

- ✓ Equipment productivity … equipment condition is maintained in accordance with design specifications and operating at peak performance.
- ✓ Fleet production … equipment breakdowns and resultant unplanned repairs are held to a minimum resulting in a reduction in failure frequency and downtime.
- ✓ Operating costs … component lives are optimized and contingent damage is avoided.
- ✓ Labor efficiency … manpower is deployed in a proactive (planned) rather than reactive (crisis “management”) manner.

- ✓ Safety … critical systems such as steering and brakes are performing to design specifications.

As a result, cost per ton will be optimized on a long-term basis resulting in increased customer satisfaction.

## 5.0 Resource Requirements

- Human Resources:

Human resource requirements associated with the implementation and execution of the Preventive Maintenance plan include:

- ✓ Planner … defines and organizes resource & time estimates to facilitate effective, efficient Preventive Maintenance execution.
  - ➢ Ensures that PM activities are aligned with the PM strategy.
  - ➢ Defines, organizes & procures facilities, manpower, parts, tooling, forms, checklists, etc. for PM.
  - ➢ Establishes priorities for PM execution.
  - ➢ Communicates and obtains approval for the PM plan from the mine Operations Department.
  - ➢ Communicates the PM plan to the Parts Department and other external resources.
- ✓ Fleet Analyst (Reliability Engineer) … provides technical assistance to the Planner related to PM content and execution.
  - ➢ Defines PM checklists for each model of equipment and provides them to the Planner.
  - ➢ Ensures that PM checklists are reflective of current manufacturers' guidelines and recommendations.
  - ➢ Updates PM checklists to comply with equipment age, application, and condition.
  - ➢ Revises PM checklists to address the key issues.
  - ➢ Follows-up / analyzes PM inspection and performance test results.
- ✓ Parts (central parts warehouse) … to support on-site Parts Department operations with timely PM parts delivery.
  - ➢ Assembles PM parts kits (mandatory, exchange, and optional) for shipment to site.
  - ➢ Assembles SOS kits for shipment to site.
  - ➢ Guarantees PM parts availability at the site.
  - ➢ Identifies PM part number changes & revise PM parts lists accordingly.
- ✓ Parts (on-site parts warehouse) … to support the Preventive Maintenance process with timely delivery of quality parts.
  - ➢ Receives PM parts kits from the central parts warehouse.
  - ➢ Verifies that the PM parts kits received agree with the PM program.
  - ➢ Stores the PM parts kits in a clean, secure location.
  - ➢ Delivers PM parts kits to the assigned location in the PM bay prior to the arrival of the equipment.

- Re-conditions the exchange PM parts kits and replenishes the optional PM parts kits.
- Retrieves the SOS bottles for shipment to the oil lab.

✓ Scheduler … translates the Preventive Maintenance plan into an actionable schedule.

- Schedules PM's in accordance with the PM plan and strategy and resource availability.
- Publishes / distributes the PM program to all appropriate areas.
- Updates the PM program to comply with variations in equipment utilization.
- Uses the "Window of Opportunity" presented by PM as well as planning goals for PM to schedule backlogged repairs in conjunction with PM.

✓ Wash Bay Team … cleans the equipment in a manner that promotes effective, efficient PM execution.

- Adheres to established procedures for equipment washing … use high-flow water cannons and smaller, portable devices as appropriate.
- Uses steam / detergent to clean areas that will be worked on during the PM.
- Avoids damage to equipment by directing water away from sensitive areas.
- Assures machine cleanliness quality for PM.

✓ Technicians … perform Preventive Maintenance activities in accordance with the guidelines and procedures outlined in the PM plan.

- Meet / exceed the goals for PM quality ($MTBS_{after\ PM}$) and efficiency ($MTTR_{PM}$).
- Document results; complete PM checklists as you go.
- Provide complete, accurate information on checklists.
- Execute all tasks included in the plan; backlogs, etc.
- Pay attention to details; use inspections, performance tests and other Condition Monitoring techniques to identify defects or potential problems.
- Correct defects immediately if possible; enter backlogs for those that cannot be addressed.

✓ PM Supervisor … directs, supervises and audits all PM tasks in a manner that enhances equipment performance and reliability.

- Ensures compliance with all PM procedures.
- Promotes effective, efficient PM execution.
- Ensures safe execution of PM procedures.
- Identifies PM resource limitations & seek solutions.
- Defines PM training needs and deficiencies.
- Verifies that all PM data integrity; ensures that PM recordkeeping is accurate and complete.
- Audits PM process to guarantee the quality of PM execution.

• Logistical Resources:

Logistical resource requirements associated with the execution of the Preventive Maintenance plan include:

✓ Facilities

- Wash Bay(s) …
  - Sized to dimensions of largest equipment.
  - Quantity sufficient to minimize delays.
  - Equipped to remove mud, grease and debris … high-volume water cannons, high-pressure guns, steam, de-greaser for selective cleaning.
  - Designed for access … platforms & ladders.
  - Drainage (water & debris), water recycling capability, oil / grease / water separation capability provided.
- PM Bay(s) …
  - Dedicated PM bay(s) / facilities.
  - Sized to dimensions of largest equipment.
  - Quantity sufficient to minimize delays.
  - Compressed air, water, & electricity supplies available.
  - Fluids handling … evacuation, delivery (filtered, metered and quick disconnects) and spillage clean-up capability.
  - Work environment … well lighted, portable lighting, doors for dust / contamination control, quality work surface, exhaust / smoke control.

✓ Tooling

- Dedicated hand tools, tool boxes, & work carts.
- Dedicated special tooling … pneumatic tools, filter cutting device, oil extraction (SOS), etc.
- Dedicated testing / diagnostic instrumentation … pressure / temperature / speed packages, particle counter, etc.
- Dedicated service / technical literature.
- Personal safety equipment.

✓ Equipment

- Personal safety and first aid provisions.
- Workbenches equipped with vices.
- Workstation … writing surface w/ computer access.
- Secure cabinetry for PM tooling & instrumentation, literature, PM and repair parts receiving / storage.
- Cleaning materials … parts washing / solvent tank, shop vacuum, shop towels, rags, soap, etc.
- Filter cart (“kidney looping”), lifting devices, chains / slings, hydraulic jacks and stands, ladders and platforms, wheel chocks, ...
- Workbenches equipped with vices.
- Fire protection / extinguishing devices.

✓ Information

- Weekly PM schedule.
- PM performance bulletin board.
- Work orders for PM and repairs (backlogs, etc) to be performed during PM.

- PM checklists specific to each model / equipment.
- Active backlog list for equipment to be maintained.
- Gantt charts for all activities performed during PM.
- Parts consists lists for PM and repairs to be performed during the PM.
- Special instructions, etc.

Significant implementation costs include logistical resources (primarily PM / wash bays and the associated tooling and equipment) and the time and expertise invested in supporting material content development (checklists, procedures, guidelines, etc.).

## 6.0 Supporting Attachments

- Cat Global Mining Equipment Management Process Map (Preventive Maintenance)
- "*Metrics (KPI's) to Assess Process Performance*" - Microsoft Word Document

## 7.0 Related Best Practices

- Goals & Objectives for Preventive Maintenance (ref. 1006 - 3.1 - 1022)
- Improved PM Operations Using Specialized Bays (ref. 0607 - 2.9 - 1081)
- Utilizing VIMS Data for Effective Planned Maintenance (ref.1006 -2.15 -1027)

## 8.0 Acknowledgements

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